

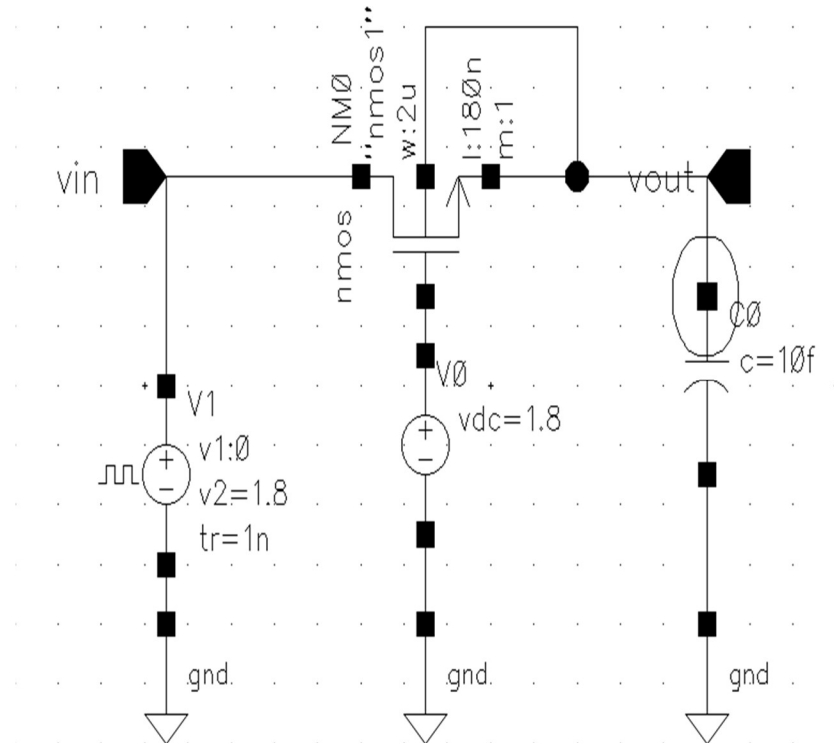
Course Code	EC24P
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Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	26 th September 2025
Date of Submission	3 th October 2025

EXPERIMENT NO.9

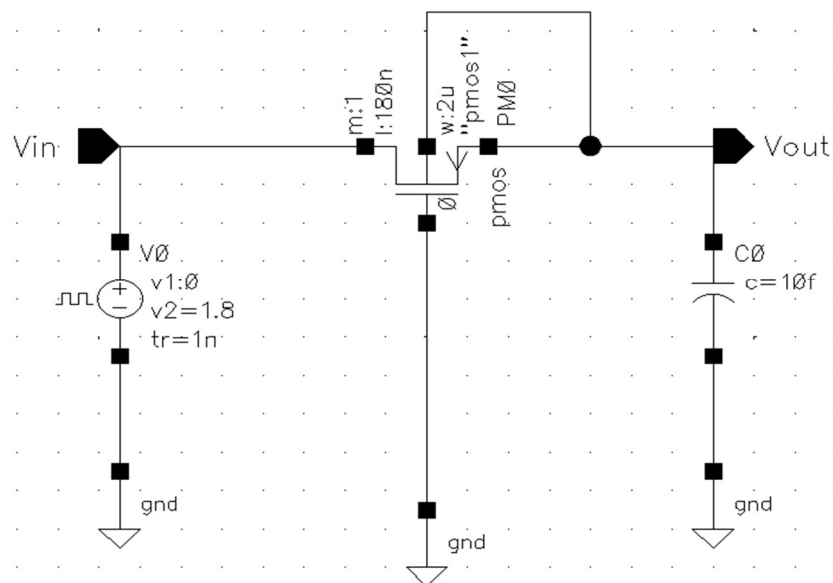
Estimated time to complete this experiment: 02 hours.
Objective: To simulate and find the rise time and fall time of CMOS inverter using gpdk180.
CO to be achieved: CO2.
Expected Outcome of Experiment: • Students will learn cadence virtuoso. • Students will be able to find rise time and fall time of asymmetric CMOS inverter. • Students will be able to find propagation delay time of CMOS Inverter.
Pre Lab/ Prior Concept: Knowledge of working of MOSFET and CMOS inverter transient analysis.
Apparatus/TOOL: • PC • CADENCE VIRTUOSO.

Circuit Diagram:

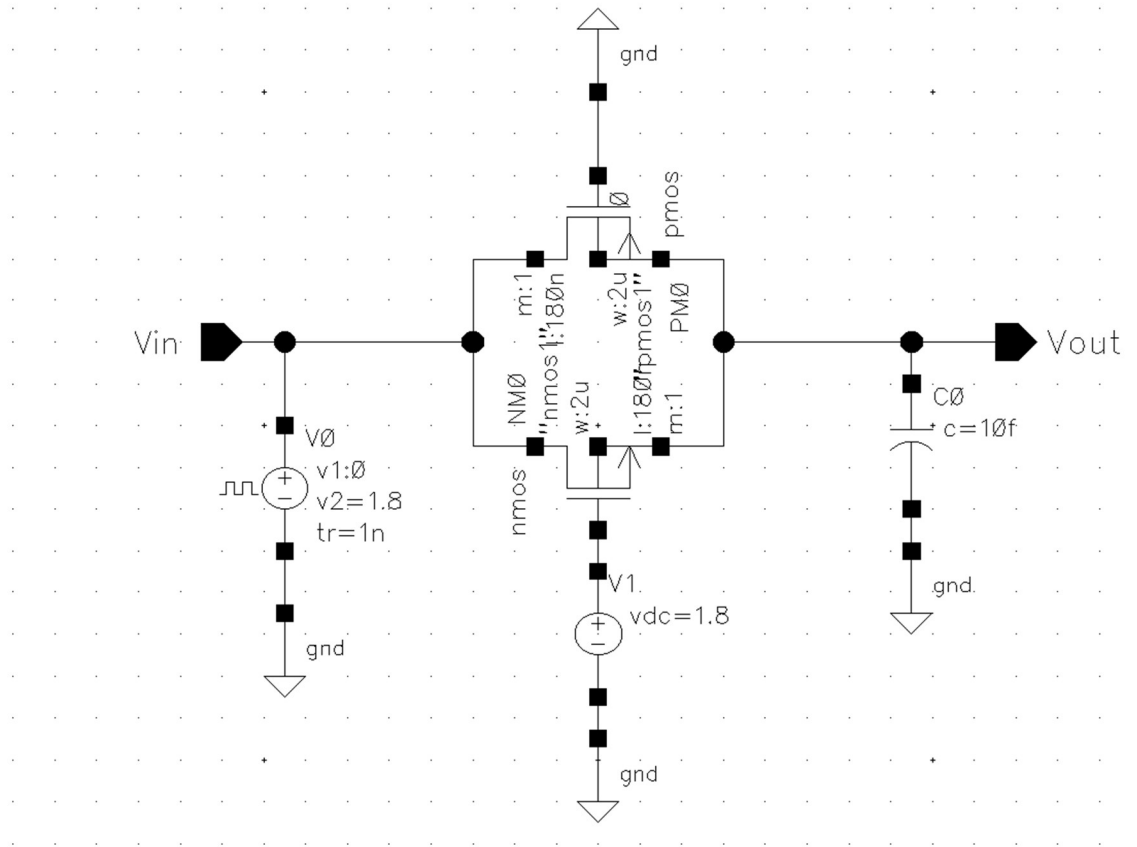
NMOS



PMOS



TG



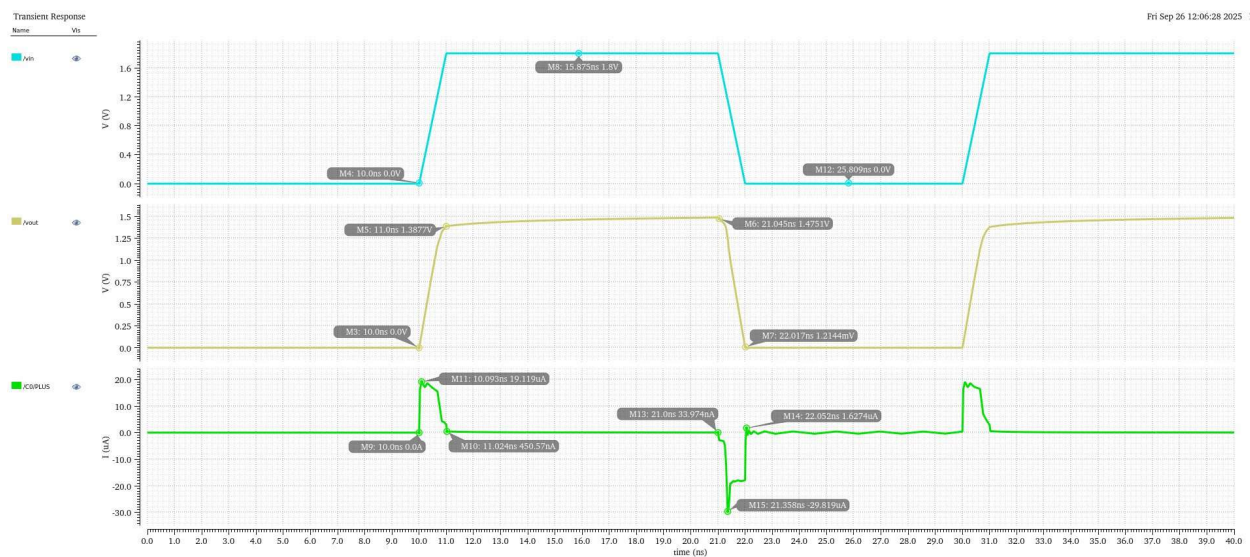
5) Procedure.

- **Open Cadence Virtuoso Tool in Red-Hat Linux**
- `csh`
`source /home/install/cshrc`
`virtuoso`
- **Create Library and Cell View**
- Open the *Log Window* from the terminal.
- Click **New → Library**, and attach the **gpd180 technology library**.
- Again click **New → Cell View** to start circuit design.
- The **Virtuoso Schematic Editor** window will open.
- **Circuit Setup – CMOS Inverter for Characterization**
- From **Add Instance**, select:
 - **NMOS transistor with channel length (L) = 180 nm and width (W) = 2 μm.**

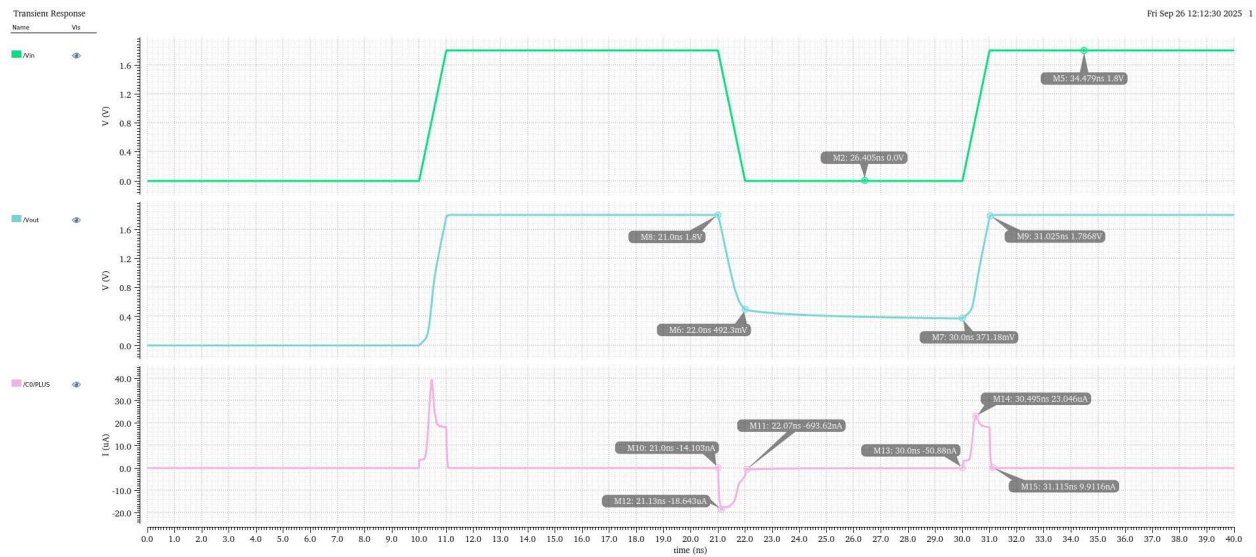
- PMOS transistor with channel length (L) = 180 nm and width (W) = 2 μm .
- Connect them to form the standard CMOS inverter circuit with VDD = 1.8 V.
- **Input Signal Setup (VPULSE source):**
- V1 = 0 V
- V2 = 1.8 V
- Delay time = 10 ns
- Rise time = 1 ns
- Fall time = 1 ns
- Pulse width = 10 ns
- Period = 20 ns
- **Perform Characterization – Effect of Channel Length Variation**
- Run **Transient Analysis** to observe input and output waveforms.
- Gradually **increase the channel length of PMOS and NMOS** (e.g., 180 nm $\rightarrow$ 2 μm $\rightarrow$ 4 μm).
- For each case, plot **input vs. output** and measure:
 - **Trise** (rise time)
 - **Tfall** (fall time)

Results:

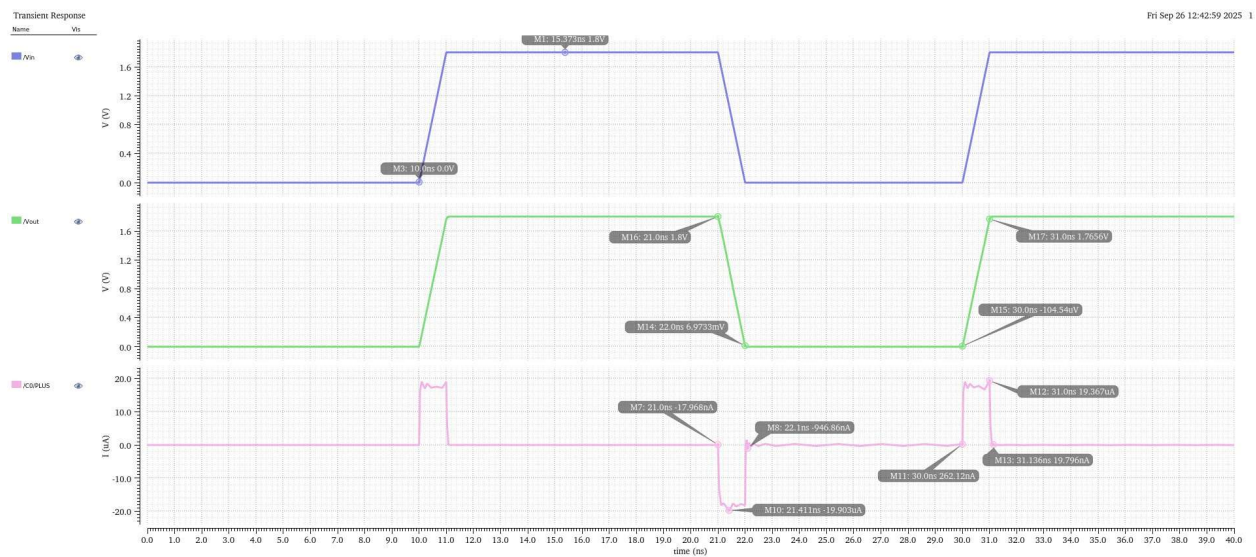
NMOS



PMOS



TG



Observation.

NMOS	
Trise	1ns
Tfall	0.972ns
Charging time	1.024ns
Discharging time	1.052ns

PMOS	
Trise	1ns
Tfall	1.025ns
Charging time	1.115ns
Discharging time	1.07ns

TG	
Trise	1ns
Tfall	1ns
Charging time	1.136ns
Discharging time	1.1ns

Conclusion:

PMOS Transistor:

- As the channel length increases, the rise time (Trise) decreases, meaning the PMOS transistor switches faster.
- The fall time (Tfall) remains almost constant across different channel lengths.

NMOS Transistor:

- Similar to PMOS, the rise time (Trise) decreases slightly with an increase in channel length.
- The fall time (Tfall) decreases significantly as the channel length increases, especially at 4u.

Increasing the channel length of both PMOS and NMOS transistors seems to lead to a slight reduction in rise times, but the fall times are more sensitive to the changes in NMOS transistors, with a sharper decrease observed at the 4u length.

Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.